

TANVIR FATIMA NAIK BUKHT

Computer Vision & Deep Learning Researcher

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Profile

Lecturer, Department of Computer Games Development (CGD), Faculty of Computing & AI, Air University, Islamabad, and PhD in Computer Science (Air University, Islamabad, completed May 2026), researching computer vision, cyber security, IoT, image processing machine learning and deep learning. Publication record spanning human activity/interaction recognition, medical image analysis, sensor/data fusion, and applied machine learning (20 journal articles, 8 conference papers; cited 818+ times per Google Scholar, Research gate). Experienced in dataset development, feature-fusion framework design, and benchmarking deep neural network architectures, with additional background in cybersecurity, QEC and IT administration.

Education

PhD, Computer Science - Air University, Islamabad, Pakistan	2021 – May 2026
MS/M.Phil, Computer Science - Institute of Southern Punjab, Multan, Pakistan	2017 – 2019
CGPA 3.02 (74%). Thesis: Analyzing the Behaviour of Cyber Attacks.	
MCS, Computer Science - Virtual University of Pakistan	2015 – 2017
CGPA 3.46 (77.19%). Project: Denial of Service Attack in Simulated Environment (Linux/C++).	
BA - Baha Uddin Zakariya University, Multan, Pakistan	2015
Intermediate - Board of Intermediate & Secondary Education, Dera Ghazi Khan	2012
Diploma in Montessori Teacher Training - Pakistan Montessori Council, Islamabad	2014

Experience

Lecturer - Department of Computer Games Development (CGD), Faculty of Computing & AI, Air University, Islamabad	2023 - Present
Instructor / Exam Coordinator - Federal Urdu University of Arts, Science & Technology, Islamabad	Sep 2018 – Mar 2019
Instructor / Exam Coordinator - Federal Urdu University of Arts, Science & Technology, Islamabad	Mar 2018 – Aug 2018
Head Teacher - Govt. Elementary School MC5, Muzaffargarh	2017 – 2019
IT Administrator / SNA - Tehsil Municipal Committee, Muzaffargarh	Jan 2015 – Jun 2017

Publications

20 peer-reviewed journal articles and 8 conference papers (2019–2026) in computer vision, human activity/interaction recognition, medical image analysis, and cybersecurity. Selected recent publications:

Almujally, N.A., Bukht, T.F.N., Alharbi, S.S., Algarni, A., Jalal, A., Park, J. (2026). Hybrid Quantum Gate Enabled CNN Framework with Optimized Features for Human-Object Detection and Recognition. Computers, Materials & Continua, 87(1), 94. <https://doi.org/10.32604/cmc.2025.072243>

Alshehri, M., Bukht, T.F.N., Alqahtani, Y., Alshahrani, A., Almujally, N.A., Jalal, A., Park, J. (2026). Complex interaction recognition via advanced multilevel feature fusion and deep learning model. PeerJ Computer Science, 12, e3514. <https://doi.org/10.7717/peerj-cs.3514>

Bukht, T.F.N., Alhaston, H.F., Alshamrari, N., Almujaally, N.A., Jalal, A. (2026). Multimodal human interaction recognition framework using multi-features and deep learning approach. International Arab Journal of Information Technology, 23(1), 129–143. <https://doi.org/10.34028/iajit/23/1/12>

Bukht, T.F.N., Alshassabi, N.S., Alhasson, H.F., Alabdullah, B., Jalal, A. (2025). A Novel Activity Pattern Recognition via Convolutional Neural Networks and Advanced Skeleton Models. Traitement du Signal, 42(4), 1889–1903. <https://doi.org/10.18280/ts.420405>

Rahman, H., Ben Aoun, N., Bukht, T.F.N., Ahmad, S., Tadeusiewicz, R., Plawiak, P., Hammad, M. (2025). Automatic liver tumor segmentation of CT and MRI volumes using ensemble ResUNet-InceptionV4 model. Information Sciences, 704, 121966. <https://doi.org/10.1016/j.ins.2025.121966>

Bukht, T.F.N., Rahman, H., Shaheen, M., Algarni, A., Almujaally, N.A., Jalal, A. (2025). A review of video-based human activity recognition: theory, methods and applications. Multimedia Tools and Applications, 84(17), 18499–18545. <https://doi.org/10.1007/s11042-024-19711-w>

Bukht, T.F.N., Alazeb, A., Al Mudawi, N., Alabdullah, B., Alnowaiser, K., Jalal, A., Liu, H. (2024). Robust Human Interaction Recognition Using Extended Kalman Filter. Computers, Materials & Continua, 81(2), 2987–3002. <https://doi.org/10.32604/cmc.2024.053547>

Rahman, H., Bukht, T.F.N., Ahmad, R., Almadhor, A., Javed, A.R., Ben Aoun, N. (2023). Efficient Breast Cancer Diagnosis from Complex Mammographic Images Using Deep Convolutional Neural Network. Computational Intelligence and Neuroscience, 2023, 7717712. <https://doi.org/10.1155/2023/7717712>

Full list of publications with links provided separately.

Professional Certificates

- Network Security Certificate - Virtual University of Pakistan
- ICSI | CNSS Certified Network Security Specialist - International Cyber Security Institute
- CCNA Cybersecurity Operations
- CCNAv7 Networks
- DDoS Attack
- Huawei Certification of Artificial Intelligence
- English Proficiency Certificate - Virtual University of Pakistan
- International Online Conference on Digitalization and Cyber Security

Additional Skills

- WordPress, Graphic Design, AutoCAD, QuickBooks
- SEO (Search Engine Optimization), E-Commerce Management
- Creative Writing, Digital Literacy
- MS Word, MS Excel, MS Outlook, PowerPoint

Languages

English, Urdu

Online Profiles

Google Scholar: scholar.google.com/citations?user=QIKqYG8AAAAJ

Faculty: <https://webdata.au.edu.pk/Pages/Faculties/FCAI/Departments/CGD/FacultyDetail.aspx?fid=2621>
dblp: dblp.org/pid/345/1223.html